

Sciama's Field Equations, the Gothic Metric, and Bottom-Up Etherodynamics

Tensor Densities, Pseudotensors, and Continuum Conservation
in a Preferred Substrate Chart

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Abstract

Sciama writes Einstein's equation, in harmonic coordinates on a Minkowski background, as a wave equation for the gothic density $\mathbf{g}^{ij} = \sqrt{-g} g^{ij}$, sourced by $\mathfrak{T}^{ij} + \theta^{ij}$ and sliced by $\partial_j \mathbf{g}^{ij} = 0$. The pair is not tensorial; the continuity law it implies is the ordinary divergence $\partial_j (\mathfrak{T}^{ij} + \theta^{ij}) = 0$. In bottom-up etherodynamics the laboratory chart is the undeformed substrate, θ^{ij} is medium stress, and the tensor/density split is the Cauchy/Piola distinction. The constitutive map of the 3 + 1D synthesis makes $\mathbf{g}^{\mu\nu}$ an algebraic composite of the optical data $(n_{ij}, \mathbf{V})$, so Sciama's pair can be written with no independent gothic field. The Landau–Lifshitz choice of $\theta[n, \mathbf{V}]$ recovers Einstein's equation after that map is inverted; the closed theory is the fixed point of the corresponding post-Minkowskian iteration. Impedance matching does not fix G_{eff} . Local rods and clocks, being excitations of the same medium, are rescaled by the constitutive fields; undetectability of that rescaling is co-deformation, and the equivalence principle is a theorem.

Contents

1 Purpose	2
2 Sciama's Maxwell Analogy	2
3 What is Tensorial, What is Not	3
4 Why Gravity Forces a Pseudotensor in Geometric GR	4
5 Tensor Densities and Local Standards	4
6 Bottom-Up Etherodynamics	5
7 What Changes under the Substrate Reading	6
8 Sciama in optical variables	11
9 Status of the Claims, Collected	15

1 Purpose

The geometric formulation of general relativity treats g_{ij} as primitive and a Minkowski background as a gauge convenience. A second, older formulation—already used by Einstein, De Donder, Papapetrou, Landau and Lifshitz, and given a physical reading by Sciama—writes the field equations as a Maxwell-like system for a tensor *density* on a flat background [1, 2, 3, 4, 5]. The two presentations are on-shell equivalent after a coordinate condition is imposed. They are not conceptually equivalent.

The purpose of this note is to keep that distinction explicit and to interpret it within a bottom-up etherodynamic programme. In that programme an all-pervading substrate is primitive; localized energy lumps (particles) and an effective acoustic metric (gravity) are derived [13, 17, 18, 19]. The gothic metric, the harmonic condition, and the gravitational energy–momentum density then describe the conversion between reference measure and current measure of that medium.

No constitutive action is varied here, and no new fundamental field equation is proposed. The note is a dictionary, together with the iterative reconstruction that makes the dictionary operational, and with the transcription of Sciama’s pair into the optical variables $(n_{ij}, \mathbf{V})$ that the 3 + 1D synthesis already treats as primitive.

2 Sciama’s Maxwell Analogy

2.1 From inertia to a tensor potential

Sciama’s 1953 paper [1] treats inertia as an inductive effect of distant matter, written with Maxwell-type equations on flat spacetime. The principle of equivalence is obtained as a consequence, not posited. The 1964 review [2] promotes the potential from a vector to a tensor and identifies the resulting theory, after a gauge condition, with Einstein’s equation.

The Maxwell prototype is

$$\square A^\mu = -J^\mu, \quad \partial_\mu A^\mu = 0 \implies \partial_\mu J^\mu = 0. \quad (1)$$

The Lorentz gauge makes the wave equation consistent with charge conservation because partial derivatives commute.

2.2 Field equation and harmonic condition

The analogous system for a tensor potential density

$$\mathfrak{g}^{ij} \equiv \sqrt{-g} g^{ij} \quad (2)$$

is Sciama’s pair (his eqs. (5)–(6) in Ref. [2])

$$\square^2 \mathfrak{g}^{ij} = -\kappa (\mathfrak{T}^{ij} + \theta^{ij}), \quad (3)$$

$$\frac{\partial \mathfrak{g}^{ij}}{\partial x^j} = 0. \quad (4)$$

Here $\square^2$ denotes the flat d’Alembertian of the background chart (historically written $\square^2$ rather than $\square$), $\mathfrak{T}^{ij} = \sqrt{-g} T^{ij}$ is the matter energy–momentum density, θ^{ij} is the gravitational/inertial-field energy–momentum *pseudotensor*, and κ is Einstein’s coupling. Papapetrou showed that (3)–(4) together are equivalent to Einstein’s equation [4, 2]. Landau and Lifshitz give the same content as an identity for a superpotential built from $\mathfrak{g}^{ij}$ [5]. In the linearized static limit the coupling matches the Einstein constant,

$$\kappa = \frac{16\pi G_{\text{eff}}}{c_0^4}, \quad (5)$$

up to the conventional factor of two that distinguishes the Landau–Lifshitz density from the Einstein tensor [5, 6]. The value of G_{eff} is discussed in Section 7.5.

2.3 How conservation follows

Partial derivatives commute with $\square^2$. Differentiating (3) and using (4) therefore yields

$$\frac{\partial}{\partial x^j} (\mathfrak{T}^{ij} + \theta^{ij}) = 0. \quad (6)$$

Equation (4) is *not* itself the conservation law. It is the harmonic (De Donder, Hilbert) coordinate condition [3]

$$\square_g x^i = \frac{1}{\sqrt{-g}} \partial_j (\sqrt{-g} g^{ij}) = 0, \quad (7)$$

equivalently $g^{jk} \Gamma_{jk}^i = 0$. Imposed on (3), it guarantees consistency with (6), exactly as the Lorentz gauge guarantees $\partial_\mu J^\mu = 0$.

3 What is Tensorial, What is Not

3.1 Transformation laws

Under a general diffeomorphism $x \mapsto x'$,

$$\mathfrak{g}'^{ij}(x') = \left| \det \frac{\partial x}{\partial x'} \right| \frac{\partial x^i}{\partial x'^k} \frac{\partial x'^j}{\partial x'^l} \mathfrak{g}^{kl}(x). \quad (8)$$

Thus $\mathfrak{g}^{ij}$ is a tensor density of weight +1, not a tensor. The same holds for $\mathfrak{T}^{ij}$. The gravitational piece θ^{ij} fails even that law: it transforms as a tensor only under *linear* changes of the background coordinates. That is the definition of a pseudotensor [5, 6].

Object	under $\text{Diff}(M)$	under Poincaré of the background
g^{ij}	tensor	tensor
$\mathfrak{g}^{ij} = \sqrt{-g} g^{ij}$	tensor density, weight +1	ordinary tensor
T^{ij}	tensor	tensor
$\mathfrak{T}^{ij} = \sqrt{-g} T^{ij}$	tensor density, weight +1	ordinary tensor
θ^{ij}	pseudotensor	tensor (inertial Cartesian chart)

3.2 Why weight +1

Three structural facts select $\sqrt{-g} g^{ij}$ rather than another power.

1. The curved-space wave equation on the coordinates themselves is $\square_g x^i = 0$ if and only if $\partial_j \mathfrak{g}^{ij} = 0$. Weight +1 makes the harmonic condition a plain divergence, which is why Sciama's Maxwell analogy is literal.
2. In inertial Cartesian coordinates of a Minkowski background one has $\sqrt{-\eta} = 1$, so $\mathfrak{g}^{ij}|_{\text{bg}} = \eta^{ij}$. The deviation from the background is, to first order, the trace-reversed perturbation of linearized gravity. That is why (3) can be written with a *flat* wave operator.
3. Weight +1 is the weight of the Einstein–Hilbert density $\sqrt{-g} R$ and of any current whose ordinary divergence integrates against d^4x . Landau–Lifshitz then form the weight-+2 combination $(-g)(T^{ij} + t_{\text{LL}}^{ij})$ so that a superpotential identity supplies (6) identically [5, 6].

3.3 Poincaré covariance of the background theory

Fix inertial Cartesian coordinates of a Minkowski background η_{ij} . The isometry group is the Poincaré group, with Jacobian $\det \Lambda = \pm 1$. For the proper orthochronous subgroup the density weight is invisible:

$$\mathfrak{g}'^{ij}(x') = \Lambda^i_k \Lambda^j_l \mathfrak{g}^{kl}(x). \quad (9)$$

Relative to that background, $\mathfrak{g}^{ij}$ is a legitimate $(2,0)$ tensor field. Equations (3)–(4) are Lorentz-covariant field equations on Minkowski space. They are *not* diffeomorphism-covariant as written, and they are not valid in an arbitrary curvilinear frame [2, 6]. The generally covariant Einstein equation $G^{ij} = \kappa T^{ij}$ is; the Maxwell-like pair is its presentation after the harmonic slice has been chosen.

“Invariant under the Poincaré group” would mean a scalar. $\mathfrak{g}^{ij}$ is not a scalar. It is Poincaré-covariant.

4 Why Gravity Forces a Pseudotensor in Geometric GR

The equivalence principle implies that at any event one may choose Riemann normal coordinates with $g_{ij} = \eta_{ij}$ and $\partial_k g_{ij} = 0$. Every candidate gravitational energy density built from first derivatives of the metric then vanishes. A genuine tensor that vanished in one frame would vanish in every frame. Hence no nonzero tensor of g_{ij} can represent a local gravitational energy–momentum density [5, 6].

Matter alone obeys the covariant law $\nabla_j T^{ij} = 0$, true in every frame but not an ordinary continuity equation: the connection terms describe exchange with the gravitational field. To obtain a continuity equation with partial derivatives one must add a non-tensorial piece θ^{ij} and work in coordinates where (4) holds. The conserved integrals constructed from $\mathfrak{T}^{ij} + \theta^{ij}$ then depend on the asymptotic inertial frame. That coordinate dependence is the same “pseudo”.

In geometric general relativity this exhausts the argument. In a theory that retains a physical background chart the conclusion is different.

5 Tensor Densities and Local Standards

The factor $\sqrt{-g}$ is the Jacobian from a local orthonormal coframe $\{\theta^a\}$ to the coordinate coframe $\{dx^\mu\}$:

$$\sqrt{-g} = |\det(e^a{}_\mu)|. \quad (10)$$

Equivalently, $\sqrt{-g} d^4x$ is the proper four-volume of the coordinate parallelepiped, measured with local rods and clocks. For a vector,

$$\partial_j(\sqrt{-g} V^j) = \sqrt{-g} \nabla_j V^j. \quad (11)$$

The ordinary divergence of the density automatically includes the expansion or contraction of the coordinate mesh relative to the local frames.

This is only a *volume* conversion. Local standards of length and duration are the full metric $g_{\mu\nu}$, ten functions. Gravitational time dilation, anisotropic stretching, and frame dragging live in those ten functions, not in $\det g$. A Weyl rescaling $g_{\mu\nu} \mapsto \Omega^2 g_{\mu\nu}$ would rescale all lengths uniformly and send $\sqrt{-g} \mapsto \Omega^4 \sqrt{-g}$; gravity is not that.

The same factor exists in flat spacetime (e.g. $\sqrt{-\eta} = r^2 \sin \theta$ in spherical coordinates). Densities track chart-volume versus metric-volume. Gravity enters only when the chart is a background inertial grid and $g_{\mu\nu} \neq \eta_{\mu\nu}$.

In the weak-field isotropic chart $ds^2 = -(1 + 2\Phi) dt^2 + (1 - 2\Phi) d\mathbf{x}^2$ one has $\sqrt{-g} \approx 1 - 2\Phi$ to first order. Relative to the Minkowski laboratory grid the proper volume of a coordinate cell does change with the Newtonian potential. That is the limited sense in which the density factor “knows about gravity.”

Eddington’s language remains useful: tensors measure intensity (per proper unit); tensor densities measure quantity per coordinate cell.

6 Bottom-Up Etherodynamics

6.1 Comparsi: continuum mechanics as the foundation

Comparsi [13] proposes that the corroborated content of physics be recovered from continuum mechanics—the first law of thermodynamics together with Newton’s second and third laws—rather than from a primitive spacetime geometry plus a primitive Hilbert space. The vacuum is treated as an aether, a continuous medium. The gravitational sector is to be written in the Landau–Lifshitz and ADM presentations of GR with physically motivated gauges; the quantum sector is to be written as nonlinear relativistic wave mechanics in $3 + 1$ dimensions, combining Barut’s self-field approach to QED with de Broglie’s double-solution programme [13, 14, 15].

Two consequences are immediate for the present note. First, the Landau–Lifshitz form is not an optional rewriting: it is the form in which GR is already a continuum theory of a medium, with ordinary divergences and a preferred class of charts. Second, particles are to be recovered as localized field configurations of that medium, not as point sources placed on an independent geometry.

6.2 Fulceri: lumps, impedance matching, effective metric

The constructive side of the same programme is developed in [16, 17, 18, 19, 21]. A non-relativistic, non-quantum, non-gravitational continuum (sine–Gordon in $1 + 1$, nonlinear Klein–Gordon with a rank-two stiffness tensor in $2 + 1$ and $3 + 1$) admits localized energy lumps. Those lumps

- remain solutions after a Lorentz boost,
- carry a rest energy $E_0 = M_0 c_0^2$ localized on a proper scale ℓ_0 ,
- possess a carrier frequency scaling as γ and an internal (clock) frequency scaling as $1/\gamma$.

Collective-coordinate reduction produces the canonical square-root particle Lagrangian; the einbein form makes the massless limit smooth [17, 21].

The rank-two stiffness (modulus) tensor of the substrate is written C^{ij} throughout this note, the standard continuum-mechanical letter for an elastic stiffness. It is the same field as the tensor denoted Θ^{ij} in the constructive notes [17, 18, 19, 20]; the renaming keeps it visually distinct from the gravito-inertial source $\theta^{\mu\nu}$. An impedance-matching condition $\rho C^{xx} = Z_0^2$ in one spatial dimension (or $\rho (\det C^{ij})^{1/d}$ in d spatial dimensions) forbids reflection while still permitting refraction. A refractive-index field n (or tensor n_{ij}) and a substrate flow $\mathbf{V}$, both defined on the background Euclidean space and Newtonian time, then determine an effective acoustic metric $g_{\mu\nu}[n, \mathbf{V}]$. The Euler–Lagrange equations of the einbein action are the affine geodesic equation of that metric [17, 18, 19]. In $2 + 1$ and $3 + 1$ dimensions the same construction reconstructs the constitutive profiles that induce the canonical Schwarzschild and Kerr geometries at vanishing radial flow [18, 19].

The equivalence principle is therefore a theorem of continuum mechanics: lumps made of the medium co-move through the same constitutive landscape, independently of their internal energy. This is the Theorem of Gravito-Inertial Equivalence of the $3 + 1$ D synthesis [19]. Emergent matter and emergent gravity are successive consequences of one substrate theory, not two independent programmes [17, 19].

The background constitutive constants of the undeformed substrate are the inertial density ρ_0 and the isotropic stiffness C_0 . They determine the reference signal speed and the mechanical impedance,

$$c_0 = \sqrt{\frac{C_0}{\rho_0}}, \quad Z_{0,\text{mech}} = \sqrt{C_0 \rho_0} = \rho_0 c_0. \quad (12)$$

Impedance matching constrains the *landscape* of (ρ, C^{ij}) so that energy lumps propagate without reflection. It does not, by itself, determine the strength with which those lumps source a deformation of the landscape. That strength is G_{eff} and is treated in Section 7.5.

6.3 Dictionary

GR / Sciamia language	etherodynamic reading
Minkowski laboratory / background chart	undeformed substrate chart (ontologically real)
g^{ij}	effective acoustic metric induced by $n, \mathbf{V}$
$\sqrt{-g}$	Jacobian from local co-moving frames of the deformed medium to the substrate cell
$\mathfrak{g}^{ij} = \sqrt{-g} g^{ij}$	flux of effective geometry per substrate cell (Piola-type density)
$\mathfrak{T}^{ij}$	stress-energy of lumps, per substrate cell
θ^{ij}	stress-energy of the substrate deformation itself
$\partial_j \mathfrak{g}^{ij} = 0$	harmonic slice on the effective metric, written with substrate partials
Poincaré group of the background	symmetry of the homogeneous, non-flowing substrate
Lorentz symmetry of particles	emergent symmetry of the lump solutions

Sciamia's pair (3)–(4) is then the infrared form of the substrate field equations in the background chart, rather than a coordinate condition imposed on a primitive geometry.

7 What Changes under the Substrate Reading

7.1 Rescaling of local standards

In geometric general relativity, local rods and clocks are not regarded as rescaled: $\sqrt{-g}$ compares a coordinate cell to a local tetrad. That remains the correct statement at the effective level.

In the substrate theory the lumps *are* the rods and clocks. In an impedance-matched index field the breather diameter scales as $1/\sqrt{n}$ and the internal clock frequency scales as $1/\sqrt{n}$ [16, 17, 19]. Length and duration are rescaled by the medium. A local observer cannot detect this, because the measuring instruments are made of the same medium and co-deform. Undetectability of the rescaling *is* the equivalence principle, now derived rather than postulated [7, 19].

Thus:

- effective geometry: local Minkowski frames, no detectable rescaling;
- substrate: standards *are* rescaled by n and $\mathbf{V}$; undetectability is co-deformation.

$\sqrt{-g}$ remains only the volume Jacobian of that co-deformation. The full stretching of time versus space sits in $g_{ij}[n, \mathbf{V}]$. The gothic metric keeps both pieces and reduces to η^{ij} when $n = 1$ and $\mathbf{V} = 0$.

7.2 Densities as Piola stresses

Continuum mechanics distinguishes Cauchy stress (force per current area) from first Piola–Kirchhoff stress (force per reference area). The tensor/density split is that distinction.

- T^{ij} is effective stress referred to proper volume in the acoustic geometry (Cauchy-like).
- $\mathfrak{T}^{ij} = \sqrt{-g} T^{ij}$ is the same stress referred to the substrate coordinate cell (Piola-like).

One integrates $\mathfrak{T}^{ij} d^4 x_{\text{lab}}$ because the laboratory cell is the reference cell of the ether. The identity $\partial_j(\sqrt{-g} V^j) = \sqrt{-g} \nabla_j V^j$ says that the ordinary divergence in the substrate chart already includes the expansion of the deformed medium relative to the reference mesh. That is why Sciamia can write conservation with ∂_j .

7.3 θ^{ij} as constitutive stress

In GR, θ^{ij} is a pseudotensor because gravitational energy cannot be a tensor of g_{ij} . In etherodynamics the paradox arises only after the substrate has been discarded.

θ^{ij} is the stress–energy of the constitutive deformation: gradients of n , kinetic energy of $\mathbf{V}$, elastic energy stored in C^{ij} . In the substrate chart it is an ordinary field tensor (or weight-+1 density) of $(\rho, C^{ij}, \mathbf{V})$. It fails to be a tensor of the *effective* metric for the same reason phonon stress is not a tensor of the acoustic metric it helps to define: the geometry is made of the field whose energy one is localizing.

Local vanishing of θ^{ij} in Riemann normal coordinates of g_{ij} is the statement that, at one event, one can co-move with the medium so that first gradients of n and $\mathbf{V}$ drop out. That is a local rest frame of the ether, not a demonstration that the medium energy is fictitious. In the laboratory chart, θ^{ij} has the same status as the elastic energy of a strained solid.

Sciama’s inclusion of θ^{ij} on the right-hand side of (3) is Machian in the original sense [1, 2]: the inertial/gravitational field is sourced by matter *and* by the field energy of the medium.

Kinematic ingredients. The acoustic tensor, the refractive-index tensor and the constitutive map of the 3 + 1D synthesis [19] are

$$A^{ij} = \frac{C^{ij}}{\rho}, \quad n_{ij} = c_0 (A^{-1/2})_{ij}, \quad \bar{n} = (\det n)^{1/3}, \quad (13)$$

$$g_{00} = -\frac{1}{\bar{n}} + n_{ij} \frac{V^i V^j}{c_0^2}, \quad g_{0i} = n_{ij} \frac{V^j}{c_0}, \quad g_{ij} = n_{ij}. \quad (14)$$

The gothic density $\mathbf{g}^{\mu\nu} = \sqrt{-g} g^{\mu\nu}$ is therefore a composite field of $(\rho, C^{ij}, \mathbf{V})$.

Mechanical ansatz. A Piola-type medium stress in the laboratory chart, organized by tensor rank, reads

$$\theta^{00} = \frac{1}{2} \rho |\mathbf{V}|^2 + U_{\text{el}}(\rho - \rho_0, C^{ij} - C_0 \delta^{ij}) + \frac{\xi^2}{2\rho_0} |\nabla \rho|^2 + \dots, \quad (15)$$

$$\theta^{0i} = \rho c_0 V^i, \quad (16)$$

$$\theta^{ij} = \rho V^i V^j + \sigma_{\text{el}}^{ij}[\rho, C^{ij}] + \text{gradient stresses}. \quad (17)$$

Here U_{el} is the stored elastic density of the stiffness deviation and σ_{el}^{ij} is the corresponding Piola stress; ξ is a weakly constrained gradient length. In the static isotropic weak-field limit the energy density must reduce to the Newtonian field energy

$$\theta^{00} \longrightarrow -\frac{|\nabla \Phi|^2}{8\pi G_{\text{eff}}}, \quad \Phi = \frac{c_0^2}{2} (1 - \bar{n}). \quad (18)$$

Equation (18) constrains the quadratic part of U_{el} once $\bar{n} = \bar{n}[\rho, C^{ij}]$ is substituted.

Canonical identification. Once $\mathbf{g}^{\mu\nu}[\rho, C^{ij}, \mathbf{V}]$ is given, the Papapetrou / Landau–Lifshitz quadratic form in $\partial \mathbf{g}$ is $\theta^{\mu\nu}$, up to a superpotential that does not change the conserved integrals [4, 5]. In the Landau–Lifshitz presentation

$$(-g)(T^{\mu\nu} + t_{\text{LL}}^{\mu\nu}) = \partial_\alpha \partial_\beta H^{\mu\alpha\nu\beta}, \quad (19)$$

with H built from $\mathbf{g}$ alone. Evaluating $t_{\text{LL}}^{\mu\nu}$ on the constitutive map (14) yields a definite function

$$\theta^{\mu\nu} = \theta^{\mu\nu}(\rho, C^{ij}, \mathbf{V}, \partial \rho, \partial C^{kl}, \partial \mathbf{V}). \quad (20)$$

That function is a tensor under Poincaré transformations of the laboratory chart, a pseudotensor under diffeomorphisms of the effective metric, reduces to (18) in the weak static limit, and makes (6) an identity whenever (3) holds.

The practical construction is therefore: insert (14) into $\mathbf{g}^{\mu\nu}$; compute the Landau–Lifshitz (or Papapetrou) expression; linearize about $(\rho_0, C_0, \mathbf{V} = 0)$ and match θ^{00} to (18); read off the quadratic piece of U_{el} . The exact Landau–Lifshitz form is retained as the nonlinear definition. The matching step determines G_{eff} only after an additional scale has been chosen (Section 7.5).

7.4 Origin of the flat wave operator

Equations (3)–(4) are valid in the background laboratory frame because that frame is the chart of the medium. The left-hand side and the right-hand side of (3) have distinct mechanical origins.

Why a wave equation. Linear signals on the undeformed substrate obey the flat d’Alembertian of the laboratory chart,

$$\square_0 = \frac{1}{c_0^2} \partial_t^2 - \nabla^2. \quad (21)$$

Any slow, long-wavelength deformation of the constitutive fields is a modulation of that same medium. The potential that encodes the deformation must therefore propagate, or relax, with a differential operator built from $\square_0$.

Why the gothic variable. Collective-coordinate reduction already supplies $g_{\mu\nu} = g_{\mu\nu}[n_{ij}, \mathbf{V}]$. The object that reduces to η^{ij} when $n = 1$ and $\mathbf{V} = 0$, and whose ordinary divergence is the harmonic condition, is $\mathbf{g}^{ij}$. To first order in the laboratory chart

$$\mathbf{g}^{\mu\nu} = \eta^{\mu\nu} - \bar{h}^{\mu\nu} + O(h^2), \quad (22)$$

so $\square_0 \mathbf{g}$ is the trace-reversed wave operator of linearized gravity. Curvature and nonlinear kinematics have been moved onto the constitutive map and onto the right-hand side. This is the Papapetrou / Landau–Lifshitz rearrangement, read as the elasticity of the ether written in the ether’s own chart.

Why the source is the total stress. Newton’s third law applied to the continuum requires that a lump, being a localized excitation of the same medium, source a deformation of $(\rho, C^{ij}, \mathbf{V})$, and that the stored elastic energy and the kinetic energy of $\mathbf{V}$ source further deformation. If one sourced only with $\mathfrak{T}^{ij}$, the identity

$$\partial_j \square_0 \mathbf{g}^{ij} = \square_0 \partial_j \mathbf{g}^{ij} = 0 \quad (23)$$

would force $\partial_j \mathfrak{T}^{ij} = 0$ in the harmonic slice, which is false: lumps exchange energy–momentum with the deformation field. The missing piece is θ^{ij} . The Maxwell analogy is now mechanical. The Lorentz gauge implies charge conservation; the harmonic slice implies conservation of lump stress plus medium stress.

Why the coefficient is κ . In the weak-field static limit the constitutive map gives $\bar{n} = 1 - 2\Phi/c_0^2$. Recovery of Poisson’s equation $\nabla^2 \Phi = 4\pi G_{\text{eff}} \rho_{\text{lump}}$ then fixes (5). The Newtonian potential Φ is a constitutive perturbation; the identification is the same as in linearized gravitation.

Equation (4) is the De Donder condition on the effective metric, expressed with substrate partials. If the laboratory coordinates are a material labelling of ether elements, it also slices residual label freedom after impedance matching. It is not a law of the medium in every chart; it is the slice in which the wave operator on $\mathbf{g}^{ij}$ is the flat $\square_0$ of the substrate. Nonlinearities in θ^{ij} are the self-gravity of the medium.

The continuity law (6) is ordinary local conservation of total substrate-plus-soliton energy–momentum in the reference chart. The covariant conservation $\nabla_j T^{ij} = 0$ in the effective geometry is the same fact after the volume factor and the connection of g_{ij} have been moved onto the left-hand side. The two conservation laws are equivalent because the effective geometry is built from the same constitutive fields that define θ^{ij} .

What is not derived in the present note is the step from a constitutive action for $(\rho, C^{ij}, \mathbf{V})$ to the statement that the Euler–Lagrange equation of that action, expressed in terms of $\mathbf{g}$, coincides

with Einstein's tensor plus a superpotential. That identification is assumed at the infrared level and is listed among the open points.

7.5 The effective coupling

The kinematic impedance (12) and the matching condition $\rho (\det C^{ij})^{1/3} = Z_{0,\text{mech}}^2$ do not determine G_{eff} . The former sets the wave resistance of the undeformed substrate; the latter forbids reflection. Newton's constant has dimensions $\text{L}^3\text{M}^{-1}\text{T}^{-2}$. From (ρ_0, c_0) alone one cannot form a quantity of that dimension: an additional length ℓ_* is required, or an equivalent cosmic scale.

Elasticity of space. If the Einstein–Hilbert prefactor is identified with the substrate tension evaluated on a microscopic cell of size ℓ_* ,

$$\frac{c_0^4}{16\pi G_{\text{eff}}} \sim \varepsilon_0 \ell_*^2 = \rho_0 c_0^2 \ell_*^2 = Z_{0,\text{mech}} c_0 \ell_*^2 = C_0 \ell_*^2, \quad (24)$$

one obtains the estimate

$$G_{\text{eff}} \sim \frac{c_0^2}{16\pi \rho_0 \ell_*^2} = \frac{c_0^3}{16\pi Z_{0,\text{mech}} \ell_*^2}. \quad (25)$$

The impedance enters as the vacuum modulus. The length ℓ_* is the cutoff at which the continuum description ceases to apply.

Take that cutoff to be the Planck length and identify c_0 with c , G_{eff} with Newton's G . In SI units

$$\begin{aligned} c &= 2.99792458 \times 10^8 \text{ m s}^{-1}, \\ G &= 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \\ \hbar &= 1.054571817 \times 10^{-34} \text{ J s}, \\ \ell_{\text{P}} &= \sqrt{\frac{\hbar G}{c^3}} = 1.616255 \times 10^{-35} \text{ m}, \\ \rho_{\text{P}} &= \frac{c^5}{\hbar G^2} = 5.155 \times 10^{96} \text{ kg m}^{-3}. \end{aligned} \quad (26)$$

Equation (25) with $\ell_* = \ell_{\text{P}}$ then fixes the undeformed substrate density to a definite fraction of the Planck density,

$$\rho_0 = \frac{c^2}{16\pi G \ell_{\text{P}}^2} = \frac{\rho_{\text{P}}}{16\pi} = 1.026 \times 10^{95} \text{ kg m}^{-3}. \quad (27)$$

The associated vacuum modulus and impedance are

$$C_0 = \rho_0 c^2 = 9.22 \times 10^{111} \text{ Pa}, \quad Z_{0,\text{mech}} = \rho_0 c = 3.07 \times 10^{103} \text{ kg m}^{-2} \text{ s}^{-1}. \quad (28)$$

A Planckian cutoff together with a rest density of order ρ_{P} therefore reproduces G up to the conventional factor 16π in (25). This is the Sakharov counting [12]. It is not an independent prediction of G , since ℓ_{P} already contains G . It is a consistency condition: if the continuum description is cut off at the Planck length, the undeformed density must be of order $\rho_{\text{P}}/16\pi$. An $O(1)$ change in the conventional factor (for example 8π instead of 16π) only rescales ρ_0 by that factor.

Without an independent determination of ℓ_* , equation (25) remains a dimensional relation rather than a prediction. Choosing $\ell_* = \ell_{\text{P}}$ converts it into a statement about ρ_0 .

Sciama–Mach route. Sciama's 1953 inductive argument requires that the cosmic inertial potential be of order c_0^2 [1],

$$\frac{G_{\text{eff}} M_{\text{U}}}{R_{\text{H}}} \sim c_0^2 \quad \Rightarrow \quad G_{\text{eff}} \sim \frac{c_0^2 R_{\text{H}}}{M_{\text{U}}}. \quad (29)$$

The additional scale is cosmological. Impedance does not appear directly; it would enter only if the inductive coupling itself were derived from $Z_{0,\text{mech}}$.

Operational definition. In the linearized map,

$$\bar{n} \approx 1 - \frac{2\Phi}{c_0^2}, \quad \nabla^2 \Phi = 4\pi G_{\text{eff}} \rho_{\text{lump}}. \quad (30)$$

Thus G_{eff} is the coefficient in the back-reaction of a lump on (ρ, C^{ij}) . That coefficient belongs to the dynamical equation for the constitutive fields, not to the kinematic matching condition. Until that equation is written, G_{eff} remains a phenomenological constant of the infrared theory, related to the vacuum modulus by (25) or (29).

7.6 Iterative construction of the self-coupled theory

The nonlinear content of (3) can be reconstructed order by order from the linear theory. Write

$$\mathfrak{g}^{\mu\nu} = \eta^{\mu\nu} + \mathfrak{h}^{\mu\nu} \quad (31)$$

and impose the harmonic slice $\partial_\nu \mathfrak{h}^{\mu\nu} = 0$ at every order. The laboratory wave operator is inverted by the flat retarded Green function,

$$\mathfrak{h}^{\mu\nu} = -\kappa \square_0^{-1} \tau^{\mu\nu}, \quad \tau^{\mu\nu} = \mathfrak{T}^{\mu\nu} + \theta^{\mu\nu}. \quad (32)$$

The medium stress θ is at least quadratic in $\mathfrak{h}$. The lump stress $\mathfrak{T}$ acquires corrections once the lumps occupy a deformed constitutive background.

Post-Minkowskian recurrence. Start from a prescribed 0th-order lump configuration on the undeformed ether:

$$\mathfrak{g}_{(0)}^{\mu\nu} = \eta^{\mu\nu}, \quad \theta_{(0)}^{\mu\nu} = 0, \quad \mathfrak{T}_{(0)}^{\mu\nu} = \text{SEM of the lumps on } (\rho_0, C_0, \mathbf{V} = 0). \quad (33)$$

The first correction is linearized gravitation,

$$\square_0 \mathfrak{h}_{(1)}^{\mu\nu} = -\kappa \mathfrak{T}_{(0)}^{\mu\nu}, \quad \mathfrak{g}_{(1)} = \eta + \mathfrak{h}_{(1)}. \quad (34)$$

The medium stress at the first nontrivial order is the Landau–Lifshitz (or Papapetrou) quadratic form evaluated on $\mathfrak{h}_{(1)}$,

$$\theta_{(2)}^{\mu\nu} = \theta_{\text{LL}}[\partial \mathfrak{h}_{(1)}]. \quad (35)$$

At the same time the lumps are recomputed on the impedance-matched background $(n_{(1)}, \mathbf{V}_{(1)})$ implied by $\mathfrak{g}_{(1)}$,

$$\mathfrak{T}_{(1)}^{\mu\nu} = \mathfrak{T}[\text{lumps on } n_{(1)}, \mathbf{V}_{(1)}]. \quad (36)$$

If $\mathfrak{T}_{(0)}$ is counted as order zero, then $\mathfrak{h}_{(1)}$ is of order G_{eff} and $\theta_{(2)}$ is of order G_{eff}^2 . The next gothic correction is

$$\square_0 \mathfrak{h}_{(2)}^{\mu\nu} = -\kappa (\mathfrak{T}_{(0)}^{\mu\nu} + \mathfrak{T}_{(1)}^{\mu\nu} + \theta_{(2)}^{\mu\nu}). \quad (37)$$

At order n ,

$$\square_0 \mathfrak{h}_{(n)}^{\mu\nu} = -\kappa \left(\sum_{k=0}^{n-1} \mathfrak{T}_{(k)}^{\mu\nu} + \sum_{k=2}^n \theta_{(k)}^{\mu\nu} \right), \quad (38)$$

where $\theta_{(k)}$ is the piece of θ_{LL} of homogeneity k in $(\mathfrak{h}_{(1)}, \dots, \mathfrak{h}_{(k-1)})$, and $\mathfrak{T}_{(k)}$ is the k th correction to the lump SEM arising from motion and internal structure in $\mathfrak{g}_{(n-1)}$. The harmonic condition is preserved by the retarded integral provided the source is conserved, which it is if θ is the Landau–Lifshitz object. If the series converges, the limit $(\mathfrak{g}, \mathfrak{T}, \theta)$ satisfies (3)–(4).

Constitutive update at each order. The primitive variables of the etherodynamic theory are not the metric components but $(\rho, C^{ij}, \mathbf{V})$. Each iteration therefore consists of four operations:

1. invert (14) on $\mathfrak{g}_{(n)}$ to obtain $n_{ij}^{(n)}$, $\mathbf{V}^{(n)}$, $\rho^{(n)}$ and $C_{ij}^{(n)}$, imposing impedance matching;
2. recompute the lumps on that background, yielding $\mathfrak{T}_{(n)}$;
3. evaluate the medium stress on that background, yielding $\theta_{(n+1)}$;
4. insert $\mathfrak{T} + \theta$ into the flat wave operator, yielding $\mathfrak{g}_{(n+1)}$.

Two distinct back-reactions run in the loop. The geodesic, or co-deformation, back-reaction is the motion of the lumps through n and $\mathbf{V}$; it is already controlled by the einbein reduction [17, 21, 19]. The constitutive back-reaction is the response of $(\rho, C^{ij}, \mathbf{V})$ themselves to $\mathfrak{T} + \theta$. That response law is the missing field equation. The iteration assumes it; it does not derive it.

Relation to the spin-2 self-coupling. The same recurrence, written for a massless spin-2 field on flat space whose source is required to be the total conserved energy–momentum of field plus matter, is the Gupta–Feynman–Thirring–Deser construction [8, 9, 10, 11]. The unique local completion, up to a cosmological term, is Einstein’s equation in harmonic coordinates. Sciama’s θ on the right-hand side of (3) is that completion written at once rather than order by order.

The iteration therefore does not invent the infrared equation. It shows how the equation is generated from linearized theory plus the demand of local conservation in the substrate chart. What it does not supply is the reason why the long-wavelength excitation of the ether is a massless spin-2 field, nor the value of G_{eff} . Those questions belong to the constitutive action.

Convergence is not guaranteed. Strong-field, horizon, and highly nonlinear constitutive regimes may leave the post-Minkowskian disk of convergence; that limitation is the same as in ordinary post-Newtonian theory. Causality in the substrate chart requires that $\square_0^{-1}$ be the retarded inverse. Constraint (elliptic) pieces appear only after a $3 + 1$ split and are the analogue of the Newtonian potential at each order.

7.7 Poincaré covariance of the substrate theory

Under Poincaré transformations of the substrate chart, $\mathfrak{g}^{ij}$ transforms as an ordinary tensor. That is now a physical statement: the homogeneous ether is Poincaré-symmetric (or Euclidean-plus-time symmetric, with Lorentz symmetry recovered in the lump sector [16, 17]).

General diffeomorphisms of the effective spacetime mix physical deformation of the medium with a mere relabelling. In the absence of a background they appear as symmetries; in the presence of a background they do not. Equations (3)–(4) need not be Diff-covariant as written. The infrared theory is a Poincaré-covariant field theory of $\mathfrak{g}^{ij}$ on the ether, equivalent on-shell to Einstein’s equation in harmonic coordinates.

8 Sciama in optical variables

The preceding sections treat $\mathfrak{g}^{\mu\nu}$ as the infrared unknown and $(\rho, C^{ij}, \mathbf{V})$ as the substrate primitives from which it is assembled. The $3 + 1$ D synthesis already packages (ρ, C^{ij}) into the refractive-index tensor n_{ij} [19, 20]. Once that packaging is used, $\mathfrak{g}^{\mu\nu}$ ceases to be an independent field: it is an algebraic composite of $(n_{ij}, \mathbf{V})$. The present section writes Sciama’s pair in those optical variables. This is the transcription of the dictionary into the variables evolved by the bottom-up programme.

8.1 Gothic metric as a composite of n_{ij} and $\mathbf{V}$

The constitutive map (14) is an ADM geometry on the laboratory chart, with lapse $\bar{n}^{-1/2}$, shift V^i/c_0 and spatial metric n_{ij} . Its determinant is independent of the flow,

$$\sqrt{-g} = \bar{n} = (\det n)^{1/3}. \quad (39)$$

The inverse metric and the gothic density are therefore algebraic in the optical data:

$$\mathfrak{g}^{00} = -(\det n)^{2/3}, \quad (40)$$

$$\mathfrak{g}^{0i} = (\det n)^{2/3} \frac{V^i}{c_0}, \quad (41)$$

$$\mathfrak{g}^{ij} = (\det n)^{1/3} (n^{-1})^{ij} - (\det n)^{2/3} \frac{V^i V^j}{c_0^2}. \quad (42)$$

The inverse map is equally algebraic. From $\mathfrak{g}^{00} < 0$,

$$\bar{n} = \sqrt{-\mathfrak{g}^{00}}, \quad \frac{V^i}{c_0} = \frac{\mathfrak{g}^{0i}}{-\mathfrak{g}^{00}}, \quad (43)$$

$$n^{ij} = \frac{1}{\sqrt{-\mathfrak{g}^{00}}} \left(\mathfrak{g}^{ij} - \frac{\mathfrak{g}^{0i} \mathfrak{g}^{0j}}{\mathfrak{g}^{00}} \right), \quad (44)$$

and n_{ij} is the matrix inverse of n^{ij} . Thus $\mathfrak{g}^{\mu\nu}$ and $(n_{ij}, \mathbf{V})$ are the same degrees of freedom in two equivalent presentations. Impedance matching further slaves the inertial density to the conformal mode,

$$\rho = \rho_0 \bar{n} = \rho_0 (\det n)^{1/3}, \quad (45)$$

so that (ρ, C^{ij}) are recovered from n_{ij} alone.

When $n_{ij} = \delta_{ij}$ and $\mathbf{V} = 0$ one has $\mathfrak{g}^{\mu\nu} = \eta^{\mu\nu}$, which is the undeformed substrate of Sections 3 and 6. Weak deviations of $\bar{n}$ from unity are the trace-reversed potential of linearized gravitation, which is why the flat laboratory wave operator of (3) can act directly on (40)–(42).

8.2 Vacuum definition of $\theta[n, \mathbf{V}]$

Set $\mathfrak{T}^{\mu\nu} = 0$ in (3). The field equation then defines the medium stress as a functional of the optical data,

$$\theta^{\mu\nu}[n, \mathbf{V}] := -\frac{1}{\kappa} \square_0 \mathfrak{g}^{\mu\nu}[n, \mathbf{V}] \quad \text{on source-free configurations.} \quad (46)$$

On a vacuum solution this is an identity. Off vacuum the same symbol $\theta[n, \mathbf{V}]$ is retained as the field part of the source: it depends on $(n_{ij}, \mathbf{V})$ and their derivatives, not on the lumps. Equation (46) is *not* to be substituted back into (3) as a functional identity for arbitrary n . That substitution cancels $\square_0 \mathfrak{g}$ on both sides and yields only $\mathfrak{T}^{\mu\nu} = 0$. The vacuum formula identifies the self-source on shell; it does not determine that source off shell.

Two readings of (46) remain available, and they are the two readings already distinguished in Section 7.3. The geometric reading takes θ to be the Landau–Lifshitz (or Papapetrou) quadratic form in $\partial \mathfrak{g}$, evaluated on (40)–(42). The mechanical reading takes θ to be the Piola stress of the substrate, built from $\rho \mathbf{V} \otimes \mathbf{V}$ and from an elastic potential $U_{\text{el}}[n, \nabla n]$. Both are functionals of $(n_{ij}, \mathbf{V})$. Only the first is guaranteed, by Deser’s uniqueness argument [11], to reproduce Einstein’s equation after the optical map is inverted.

8.3 Sciama written in $(n_{ij}, \mathbf{V}, \mathfrak{T})$

Substitute (40)–(42) into (3)–(4) and keep θ as a functional of the optical data. The gothic field disappears as an unknown:

$$\square_0 [-(\det n)^{2/3}] = -\kappa (\mathfrak{T}^{00} + \theta^{00}[n, \mathbf{V}]), \quad (47)$$

$$\square_0 \left[(\det n)^{2/3} \frac{V^i}{c_0} \right] = -\kappa (\mathfrak{T}^{0i} + \theta^{0i}[n, \mathbf{V}]), \quad (48)$$

$$\square_0 \left[(\det n)^{1/3} (n^{-1})^{ij} - (\det n)^{2/3} \frac{V^i V^j}{c_0^2} \right] = -\kappa (\mathfrak{T}^{ij} + \theta^{ij}[n, \mathbf{V}]). \quad (49)$$

The harmonic slice is the ordinary laboratory divergence of the same composites,

$$\partial_t [-(\det n)^{2/3}] + \partial_i \left[(\det n)^{2/3} \frac{V^i}{c_0} \right] = 0, \quad (50)$$

$$\partial_t \left[(\det n)^{2/3} \frac{V^i}{c_0} \right] + \partial_j \left[(\det n)^{1/3} (n^{-1})^{ij} - (\det n)^{2/3} \frac{V^i V^j}{c_0^2} \right] = 0. \quad (51)$$

Equations (47)–(51) are the infrared field equations of the substrate theory in the variables of [19]. The left-hand sides are the laboratory wave operator acting on algebraic functions of the optical data. The right-hand sides are the lump density plus the medium self-stress. No third geometric unknown is evolved.

These are the infrared equations of the bottom-up programme in the variables of [19]. The fast field u of the nonlinear Klein–Gordon continuum is defined on the laboratory chart, with $(n_{ij}, \mathbf{V})$ as coefficients; its coarse-grained stress is $\mathfrak{T}^{\mu\nu}$. The slow fields $(n_{ij}, \mathbf{V})$ obey (47)–(49). The post-Minkowskian iteration of Section 7.6 solves the same system order by order: at each step one updates n and $\mathbf{V}$ rather than $\mathbf{g}$, the two being interchangeable by (40)–(44).

Harmonic slice and continuum balances. Equations (50)–(51) have the same differential skeleton as the local conservation of energy and of momentum of the substrate, which Comparsi takes as primitive [13] and which the 3+1D synthesis writes as laboratory continuity and momentum balance for $(\rho, \mathbf{V}, \sigma_{\text{el}})$ [19]. Written with $\bar{n} = (\det n)^{1/3}$ and $\rho = \rho_0 \bar{n}$, those balances are

$$\partial_t \rho + \partial_i (\rho V^i) = 0, \quad (52)$$

$$\partial_t (\rho V^i) + \partial_j (\rho V^i V^j + \sigma_{\text{el}}^{ij}) = f_{\text{lump}}^i. \quad (53)$$

The first is continuity of $\bar{n}$. The time component (50) is, up to the conversion $x^0 = c_0 t$, continuity of $\bar{n}^2$. The spatial components (51) are conservation of the gothic momentum density $\mathbf{g}^{0i} \propto \bar{n}^2 V^i$, with flux equal to the spatial gothic metric (42). Both pairs are statements of the form $\partial_\nu J^{\mu\nu} = 0$ for a symmetric current assembled from a density and a flow. One current is the substrate mass–momentum $(\rho, \rho \mathbf{V})$. The other is the Piola flux of the inverse effective geometry, already identified as such in Section 7.

The resemblance is the optical transcription of Sciama’s Maxwell analogy [1, 2]. There the Lorentz condition $\partial_\mu A^\mu = 0$ converts $\square A^\mu = -J^\mu$ into $\partial_\mu J^\mu = 0$. Here the harmonic slice (4) converts (3) into the ordinary conservation law (6). In optical variables the same implication reads as follows: continuity of $\bar{n}^2$ is the gauge condition under which continuity of $\mathfrak{T} + \theta$ is compatible with a flat laboratory wave operator acting on (40)–(42). Equations (52)–(53) are conservation laws of the medium. Equations (50)–(51) are the coordinate slice in which those laws are consistent with (47)–(49).

The two pairs do not coincide, and the mismatch is part of the dictionary. Mass continuity conserves $\bar{n}$; the harmonic time component conserves $\bar{n}^2$. Expanding,

$$\begin{aligned} \partial_t \bar{n}^2 + \nabla \cdot (\bar{n}^2 \mathbf{V}) &= 2\bar{n} \left(\partial_t \bar{n} + \mathbf{V} \cdot \nabla \bar{n} + \frac{1}{2} \bar{n} \nabla \cdot \mathbf{V} \right), \\ \partial_t \bar{n} + \nabla \cdot (\bar{n} \mathbf{V}) &= \partial_t \bar{n} + \mathbf{V} \cdot \nabla \bar{n} + \bar{n} \nabla \cdot \mathbf{V}. \end{aligned} \quad (54)$$

The two vanish together only if $\nabla \cdot \mathbf{V} = 0$ or $\bar{n} = \text{const}$. In a compressible, optically inhomogeneous substrate they are distinct: harmonic gauge constrains a Jacobian-weighted density, because $\mathbf{g}^{00} = -\sqrt{-g} g^{00}$ already includes one power of the volume factor of Section 7. The spatial mismatch has the same origin. The Euler flux in (53) is $\rho \mathbf{V} \otimes \mathbf{V} + \sigma_{\text{el}}$. The gothic flux in (51) is $-\bar{n}^2 \mathbf{V} \otimes \mathbf{V} / c_0^2 + \bar{n} n^{-1}$. The minus sign is the ADM shift; the term $\bar{n} n^{-1}$ is the Piola image of the inverse refractive-index tensor, not the elastic stress. Setting the two fluxes equal is the constitutive identification $\theta = t_{\text{LL}}[\mathbf{g}[n, \mathbf{V}]]$ of the next subsection.

Three consequences follow for the programme.

First, a fully material harmonic chart—one in which the labels of ether elements are harmonic and ρ is conserved—exists globally only for an optically homogeneous ether. Gravity ($n \neq 1$) forces a distinction between mass continuity and gothic continuity. That is the continuum reading of the statement that (4) is a gauge condition, not a material law.

Second, in the static SSS sector ($\mathbf{V} = 0, \partial_t = 0$) the time component of the slice is empty and (51) collapses to $\partial_j(\bar{n}(n^{-1})^{ij}) = 0$. On the exact harmonic-Schwarzschild n_{ij} this is an identity, and it is the same identity as $\partial_j\theta^{ij} = 0$ in vacuum. Gauge condition and momentum conservation coincide on shell in that sector, which is why the $1/\rho^4$ Maxwell stress of Section 7.3 can be read from either.

Third, the iteration of Section 7.6 updates $(n, \mathbf{V})$ by a retarded integral of $\mathfrak{T} + \theta$. The optical harmonic slice is preserved if and only if that source is conserved. The balances that preserve (50)–(51) under iteration are therefore (6) for the total energy–momentum density, not (52) for ρ alone. This is Sciama’s consistency argument [2], expressed in the variables of the etherodynamic theory.

8.4 Plausible closures for $\theta[n, \mathbf{V}]$

A sourced theory requires a $\theta[n, \mathbf{V}]$ that is *not* identical to $-\square_0 \mathbf{g}[n, \mathbf{V}]/\kappa$. Three closures are compatible with the dictionary already in Section 7.3, and they stand in a definite logical order.

Weak isotropic mechanics. Only the conformal mode of n_{ij} is dynamical. With $\rho = \rho_0 \bar{n}$,

$$\theta^{00} = \frac{1}{2} \rho_0 \bar{n} |\mathbf{V}|^2 + \frac{c_0^4}{32\pi G_{\text{eff}}} |\nabla \log \bar{n}|^2, \quad (55)$$

$$\theta^{0i} = \rho_0 \bar{n} c_0 V^i, \quad (56)$$

$$\theta^{ij} = \rho_0 \bar{n} V^i V^j + \frac{c_0^4}{16\pi G_{\text{eff}}} \left(\partial^i \log \bar{n} \partial^j \log \bar{n} - \frac{1}{2} \delta^{ij} |\nabla \log \bar{n}|^2 \right). \quad (57)$$

The gradient piece is the Maxwell stress of the Newtonian potential $\Phi = (c_0^2/2)(1 - \bar{n}^{-1})$. On the weak SSS profile $\bar{n} = 1 + 2GM/(c_0^2 \rho)$ it reproduces the $1/\rho^4$ tidal stress of Section 7.3. This is the minimal infrared law. It does not distinguish $n_r \neq n_\perp$ and it does not reproduce the strong-field factor in θ^{00} .

Weak anisotropic mechanics. Retain the conformal pieces of (55)–(57) and add a shear energy for the trace-free optical strain

$$\tilde{n}_{ij} = n_{ij} - \bar{n} \delta_{ij}. \quad (58)$$

The shear energy density and the associated Maxwell stress are

$$U_{\text{shear}} = \frac{c_0^4}{32\pi G_{\text{eff}}} \partial_k \tilde{n}_{ij} \partial^k \tilde{n}^{ij}, \quad (59)$$

$$\sigma_{\text{shear}}^{ij} = \frac{c_0^4}{16\pi G_{\text{eff}}} \left(\partial^i \tilde{n}^{kl} \partial^j \tilde{n}_{kl} - \frac{1}{2} \delta^{ij} \partial_p \tilde{n}_{kl} \partial^p \tilde{n}^{kl} \right). \quad (60)$$

The full weak anisotropic source, written in the same pattern as (55)–(57), is then

$$\theta^{00} = \frac{1}{2} \rho_0 \bar{n} |\mathbf{V}|^2 + \frac{c_0^4}{32\pi G_{\text{eff}}} |\nabla \log \bar{n}|^2 + U_{\text{shear}}, \quad (61)$$

$$\theta^{0i} = \rho_0 \bar{n} c_0 V^i, \quad (62)$$

$$\theta^{ij} = \rho_0 \bar{n} V^i V^j + \frac{c_0^4}{16\pi G_{\text{eff}}} \left(\partial^i \log \bar{n} \partial^j \log \bar{n} - \frac{1}{2} \delta^{ij} |\nabla \log \bar{n}|^2 \right) + \sigma_{\text{shear}}^{ij}. \quad (63)$$

When $\tilde{n}_{ij} = 0$ the system reduces to (55)–(57). The shear terms are the simplest mechanical addition that can support a nonzero vacuum θ^{ij} when $\bar{n}$ is constant but n_{ij} is not. The closure remains a weak-field ansatz.

Landau–Lifshitz / Einstein closure. Evaluate the Landau–Lifshitz quadratic form on the composite (40)–(42). Let $\mathfrak{g}_{\mu\nu}$ denote the matrix inverse of $\mathfrak{g}^{\mu\nu}$,

$$\mathfrak{g}_{\mu\lambda}\mathfrak{g}^{\lambda\nu} = \delta_{\mu}^{\nu}, \quad (64)$$

so that $\mathfrak{g}_{\mu\nu} = g_{\mu\nu}/\sqrt{-g}$. All derivatives below are ordinary laboratory partials. The weight-+2 Landau–Lifshitz density is the unique quadratic polynomial in $\mathfrak{g}$ and $\partial\mathfrak{g}$ that satisfies the four conditions of Ref. [5] [6],

$$\begin{aligned} (-g)t_{\text{LL}}^{\alpha\beta} = \frac{1}{\kappa} \bigg[& \partial_{\lambda}\mathfrak{g}^{\alpha\beta}\partial_{\mu}\mathfrak{g}^{\lambda\mu} - \partial_{\lambda}\mathfrak{g}^{\alpha\lambda}\partial_{\mu}\mathfrak{g}^{\beta\mu} \\ & + \frac{1}{2}\mathfrak{g}^{\alpha\beta}\mathfrak{g}_{\lambda\mu}\partial_{\rho}\mathfrak{g}^{\lambda\nu}\partial_{\nu}\mathfrak{g}^{\mu\rho} \\ & - \mathfrak{g}^{\alpha\lambda}\mathfrak{g}_{\mu\nu}\partial_{\rho}\mathfrak{g}^{\beta\nu}\partial_{\lambda}\mathfrak{g}^{\mu\rho} - \mathfrak{g}^{\beta\lambda}\mathfrak{g}_{\mu\nu}\partial_{\rho}\mathfrak{g}^{\alpha\nu}\partial_{\lambda}\mathfrak{g}^{\mu\rho} \\ & + \mathfrak{g}_{\lambda\mu}\mathfrak{g}^{\nu\rho}\partial_{\nu}\mathfrak{g}^{\alpha\lambda}\partial_{\rho}\mathfrak{g}^{\beta\mu} \\ & + \frac{1}{8}(2\mathfrak{g}^{\alpha\lambda}\mathfrak{g}^{\beta\mu} - \mathfrak{g}^{\alpha\beta}\mathfrak{g}^{\lambda\mu})(2\mathfrak{g}_{\nu\rho}\mathfrak{g}_{\sigma\tau} - \mathfrak{g}_{\rho\sigma}\mathfrak{g}_{\nu\tau})\partial_{\lambda}\mathfrak{g}^{\nu\tau}\partial_{\mu}\mathfrak{g}^{\rho\sigma} \bigg]. \end{aligned} \quad (65)$$

The note’s source is the weight-+1 density

$$\theta^{\mu\nu}[n, \mathbf{V}] = \sqrt{-g}t_{\text{LL}}^{\mu\nu} = \frac{(-g)t_{\text{LL}}^{\mu\nu}}{\sqrt{-g}}, \quad (66)$$

evaluated on (40)–(42), with $\sqrt{-g} = \bar{n} = (\det n)^{1/3}$. Equivalently, the same object is obtained from the superpotential $H^{\alpha\mu\beta\nu} = \mathfrak{g}^{\alpha\beta}\mathfrak{g}^{\mu\nu} - \mathfrak{g}^{\alpha\nu}\mathfrak{g}^{\mu\beta}$ by subtracting the Einstein tensor from $\partial_{\mu}\partial_{\nu}H^{\alpha\mu\beta\nu}$; the second derivatives cancel and (65) remains.

This is the unique local quadratic completion that makes (47)–(51) identical to Einstein’s equation in harmonic coordinates [5, 4, 11]. The optical map inverted, one recovers $G^{\mu\nu} = \kappa T^{\mu\nu}$. The weak isotropic law (55)–(57) is the linearised isotropic sector of (66). The exact harmonic-Schwarzschild θ^{ij} of the laboratory chart is the SSS evaluation of (66).

In the bottom-up programme (66) is the infrared *target*, not the starting point. Comparisi’s insistence that the Landau–Lifshitz presentation is the physically correct form of GR [13] is the statement that (66) is the continuum stress of the medium once the long-wavelength excitation has been identified as a massless spin-2 field. A constitutive action for $(\rho, C^{ij}, \mathbf{V})$, or equivalently for $(n_{ij}, \mathbf{V})$, is still required to derive that identification and the value of G_{eff} (Section 7.5). Until that action is written, (47)–(49) with (66) are Einstein’s equation in optical variables, and (55)–(57) are their Newtonian limit expressed in the language of the substrate.

9 Status of the Claims, Collected

1. Equations (3)–(4) hold in the laboratory/substrate chart, not in a generic frame of the effective geometry. In geometric GR that is a limitation of the presentation. In etherodynamics it is a feature.
2. θ^{ij} is a pseudotensor relative to g_{ij} . Relative to the substrate it is ordinary medium stress. The “pseudo” character is what remains if the theory is written using only the effective metric. A definite identification is the Landau–Lifshitz / Papapetrou quadratic form evaluated on the constitutive map (14).
3. $\mathfrak{g}^{ij}$ is Poincaré-covariant on the Minkowski background because that background is the homogeneous ether. It is the variable that reduces to η^{ij} when the ether is unstrained and at rest, and whose ordinary divergence is the harmonic condition.

4. The passage from tensors to tensor densities accounts for the gravitational *volume* conversion from co-deformed (current) measure to laboratory (reference) measure. The full rescaling of rods and clocks is in n and $\mathbf{V}$, hence in g_{ij} . The gothic metric multiplies the two pieces; it does not replace g^{ij} by $\sqrt{-g}$.
5. Sciama's Maxwell analogy is the native infrared language of etherodynamics: wave operator of the substrate on the left, lumps plus medium self-stress on the right, gauge condition a plain background divergence. The structure is required by local conservation in the substrate chart together with the existence of a flat laboratory d'Alembertian.
6. The closed nonlinear theory is the fixed point of the post-Minkowskian iteration of Section 7.6. That iteration is the etherodynamic reading of the Gupta–Feynman–Thirring–Deser self-coupling.
7. The coupling G_{eff} is not fixed by impedance matching. It is set by the vacuum modulus $Z_{0,\text{mech}}c_0$ together with a microscopic length ℓ_* , or by a cosmic Mach scale, and remains a phenomenological constant until the constitutive back-reaction is derived.
8. The gothic metric is an algebraic composite of the optical data $(n_{ij}, \mathbf{V})$. Sciama's pair can therefore be written as (47)–(51), with no independent geometric field. The Landau–Lifshitz closure (66) makes that system identical to Einstein's equation after the optical map is inverted; the weak isotropic law (55)–(57) and the weak anisotropic law (61)–(63) are its Newtonian sectors.
9. The optical harmonic conditions (50)–(51) have the same skeleton as the substrate energy and momentum balances (52)–(53), but they conserve $\bar{n}^2$ and the gothic flux rather than ρ and σ_{el} . They are the gauge slice that makes (6) compatible with the flat wave operator; they are not themselves the material conservation laws.

The geometric theory is the infrared, diffeomorphism-covariant, background-independent presentation of the same continuum. Bottom-up etherodynamics retains the background; densities, ordinary divergences, and gravitational energy–momentum densities then record the conversion between reference measure and current measure of the medium.

Open problems

The dictionary and the iterative reconstruction constrain the infrared theory but do not replace a constitutive action. In particular:

- the explicit expansion of (66) on (40)–(42), and the constitutive potential U_{el} that reproduces it, remain to be written out;
- the residual gauge content of (50)–(51) after impedance matching has not been classified, nor has a material chart in which those conditions coincide with (52)–(53) beyond the homogeneous or static SSS sectors;
- the constitutive back-reaction—now the system (47)–(49) with a mechanical $\theta[n, \mathbf{V}]$ derived from a substrate action rather than from (66)—is the continuum origin of gravity generation and the missing input of the iteration [19];
- the length ℓ_* (or the Mach scale) that converts the vacuum modulus into G_{eff} is not derived;
- the double-solution sector—the relation between the localized u -wave (the lump) and the extended guiding wave of de Broglie and Barut [14, 15, 13]—is complementary to the gravitational reduction and is not treated here.

Acknowledgements

The interpretation of Sciama’s formulation developed here is part of a continuum-mechanical programme pursued independently by Federico Comparsi and by the author. Comparsi’s insistence that the Landau–Lifshitz presentation is the physically correct starting point, rather than an optional rewrite, is the reason the gothic metric is treated as a primitive rather than an auxiliary object.

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